Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Agentic Semantic Data Layer for Autonomous Scientific Data Discovery, Integration, and Analysis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	Culture, Medical, and Environmental based Knowledge Representation / Intelligent Convergence Technology	Software	Software Solutions
Concept of Proposed Technology	We propose the development of an agentic semantic data layer and scientific data-agent stack that enables autonomous discovery, integration, interpretation, and analysis of complex scientific data across heterogeneous data sources. The core concept is to build a software layer that continuously constructs machine-interpretable knowledge representations over structured and unstructured data, enabling intelligent agents to reason about data assets, generate analysis workflows, validate data quality, and produce explainable scientific insights. The proposed system integrates (i) automatic semantic indexing of data lakes, (ii) graph-based knowledge representations, and (iii) LLM-driven planning and tool-calling agents that operate directly on the data layer. The platform is designed as a general scientific software infrastructure applicable to data from many disciplines, including biomedical research, materials science, environmental monitoring, social science, and any domain involving large, heterogeneous datasets. We will work with multidisciplinary partners to identify the highest-impact fields to target for demonstrations. The resulting technology enables semantic level—requesting hypotheses, comparisons, and analyses—while agents autonomously locate relevant data, construct data products, assess			

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	reliability, and generate reproducible analysis pipelines.
Rationale for Joint R&D Project	Modern scientific research increasingly relies on large, heterogeneous, and continuously evolving datasets. However, current data management systems remain largely manual, fragmented, and brittle, creating bottlenecks in data discovery, integration, quality control, and reproducibility. Thus, the impact of modern AI has not been felt in scientific production-level research and discovery compared to other fields despite its immense potential. Building robust agentic data systems requires close collaboration between academic research groups with strengths in network science, semantic modeling, and LLM-based agents, and industrial partners with expertise in scalable data infrastructure and deployment. Joint R&D is essential to (i) co-design scalable semantic representations aligned with real enterprise and scientific data lakes, (ii) validate autonomous data agents in realistic industrial and scientific workflows, and (iii) transition core technologies from laboratory prototypes to deployable platforms. The collaboration enables rapid technology maturation while addressing fundamental research challenges in autonomous data reasoning and scientific AI. In other words, the goal is that industry leaders will be able to employ these scientific data agents to rapidly produce novel research and resolve it into prototypes using their existing data and expertise at a rate only possible due to AI. This goal is not achievable without joint R&D across the board.
Key Words	1. Agentic Data Systems 2. Semantic Data Layer 3. Scientific AI 4. Autonomous Data Discovery 5. Knowledge Graphs

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3–4)	End of the Project	Level (5–6)
R&D Objectives	• To design and implement an autonomous semantic data layer that builds structured knowledge representations over large scientific data lakes. • To develop LLM-empowered data agents capable of discovering data assets, generating data products, validating quality, and planning analysis workflows. • To create agentic mechanisms for data integration, schema induction, text-to-table conversion, and graph-based reasoning across heterogeneous modalities. • To demonstrate generality across multiple scientific domains, with close collaboration with industry partners to find applicable domains from the get-go. • To establish quantitative benchmarks for data utility, insight quality, and autonomous reasoning performance.			
R&D Target Specifications	• **Semantic indexing coverage** of ≥90% of ingested structured and unstructured datasets. • **Autonomous data discovery** precision ≥85%. • **Data product generation** success rate ≥80% for benchmark tasks. • **End-to-end workflow automation** reducing human intervention by ≥50%. • **Explainable reasoning traces** for ≥90% of generated analyses.			
R&D Plan	Phase 1 (Months 1–12): Semantic Layer Construction • Construct AI agent paradigms for the data layer and scientific knowledge understanding • Design knowledge representation schemas for tabular, textual, and multimodal data • Implement automatic semantic indexing and metadata extraction • Build graph-based representations and data layer tools • Text integration of all the above in one seamless platform • Apply platform to a specific domain as initial prototype Phase 2 (Months 13–24): Agentic Data and Scientific Operations • Develop data discovery, integration, and quality assessment agents • Implement text-to-table and schema induction pipelines • Integrate LLM-based planning and tool-calling mechanisms • Engage in domain-specific research tasks end-to-end (proposal, experimentation and outcomes) Phase 3 (Months 25–36): Scientific Applications and Evaluation • Apply system to multiple domains (mass spectrometry, biomedical data, social data or other domains as suggested by industry partners) • Establish benchmarks for insight quality, coverage, and reproducibility • Validate scalability and robustness in joint academic–industrial environments			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Design of semantic representations and knowledge graph architectures • Development of agentic reasoning algorithms and planning agents • Implementation of evaluation metrics for data utility and insight quality • Scientific validation across multiple research domains • Training of students and dissemination of open scientific tools
Expected R&R of Korean SME Partner	• Development of scalable data infrastructure and deployment pipelines • Integration with enterprise and scientific data lakes • System optimization, performance engineering, and testing • Industrial validation and user-driven refinement • Commercialization and productization planning

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project will deliver a general-purpose agentic data platform that transforms how scientific and industrial data are explored, integrated, and analyzed. Technological outcomes include autonomous data discovery agents, scalable semantic data layers, and explainable AI systems for scientific reasoning. The platform enables faster hypothesis generation, improved reproducibility, and significant reduction in manual data engineering effort. From a commercialization perspective, the proposed technology forms the core of next-generation data infrastructure products for scientific research institutions, industrial R&D centers, and enterprise data platforms. The semantic data layer and agentic modules can be packaged as: • A licensed software platform for enterprise and research data lakes • A cloud-based data-agent service for autonomous analytics and workflow generation • Domain-specific vertical solutions for life sciences, materials discovery, and environmental monitoring The joint R&D will enable Korean SMEs to internalize advanced agentic data technologies, accelerate product development cycles, and enter high-value global markets in scientific AI, enterprise analytics, and data engineering platforms. Expected economic impact includes the creation of new AI software products, expansion of high-skill employment, and strengthened competitiveness in international data-centric industries.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization